A LOW COST AUTOMATED MOSQUITO TRAPPING SYSTEM USING ARDUINO

MINI-PROJECT REPORT

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,

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BONAFIDE CERTIFICATE

Certified that this project report titled "**A LOW COST AUTOMATED MOSQUITO TRAPPING SYSTEM USING ARDUINO** "is the bonafide work of **Mr.R.ASWIN RAAJ (REG.No:9517202302011), Mr.S.MANI BALAJI (REG.NO:9517202302056) and Mr.V.S.THARUN (REG.NO:9517202302108)** who carried out the mini-project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Mosquitoes transmit various diseases, including malaria, dengue, and Zika virus, posing significant public health risks. This project introduces an Arduino-based mosquito trapping system that uses frequencies generated by speaker to generate sound frequencies in the range of 250 to 650 Hz, which are believed to attract mosquitoes.

At the heart of the system is an Arduino microcontroller, which allows for precise control over the frequency output. This flexibility enables users to adjust the settings based on environmental conditions and mosquito behaviour, optimizing the trapping process. This feature not only aids in reducing mosquito populations but also minimizes reliance on chemical repellents, promoting a more environmentally friendly approach to pest control.

The design of the system emphasizes user-friendliness and sustainability, making it suitable for various applications, including residential areas, gardens, and outdoor events. The automatic trapping mechanism ensures that captured mosquitoes are contained without the need for manual intervention, further enhancing convenience for users.

Controlled by an Arduino microcontroller, a speaker that emits frequencies appealing to mosquitoes. The trapping mechanism incorporates suction unit and UV lights for efficient capture. By providing an eco-friendly solution that reduces reliance on chemical repellents, this project aims to mitigate mosquito populations and address the growing threat of mosquito-borne diseases.

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CHAPTER 1

INTRODUCTION

1.1 Background on Mosquitoes

The global health crisis linked to mosquito-borne diseases, such as malaria, dengue fever, and Zika virus, presents urgent challenges for public health systems worldwide. These diseases pose significant health risks, particularly in tropical and subtropical regions, affecting millions and resulting in substantial healthcare costs and socioeconomic burdens.

Insecticides can have adverse environmental effects, contribute to pesticide resistance, and pose health risks to non-target organisms, including humans and pets. Conventional traps often lack effectiveness in diverse and dynamic environments. Thus, there is an urgent need for innovative, sustainable solutions.

This project aims to harness sound frequencies as an alternative attractant to develop a low-cost mosquito trapping system, leveraging technology to provide an eco-friendly solution to this growing problem.

By understanding and applying the behavioural cues that attract mosquitoes, this project seeks to contribute to public health efforts through innovative design and technology integration

1.2 Objective

The primary objectives of this project are to design and develop an innovative, low-cost mosquito trapping system utilizing sound frequency as a key